DATA SCIENCE FINAL PROJECT

Chicago Airbnb Pricing Analysis

What determines nightly price for a short-term Airbnb listing in Chicago?

AUDIENCE

Airbnb hosts & property investors

SCOPE

6,162 Chicago listings · Sept 2025

GOAL

Predict nightly price from listing & neighborhood features

OVERVIEW

TARGET AUDIENCE

Airbnb Hosts
Anyone hoping to list a short-term rental in Chicago who wants a starting point for optimal pricing

Property investors
Anyone evaluating which Chicago neighborhoods offer the best short-term rental returns

Airbnb Listings

Inside Airbnb · Sept 2025

6,162 active listings. Price, room type, bedrooms, bathrooms, amenities, host info.

Crime Data

Chicago Data Portal · Feb 2025–Feb 2026

Non-domestic incidents aggregated by community area — total, violent, theft, arrest rate.

Income Data

ACS Estimates · 2020–2024

Median household income for all 77 Chicago community areas.

Boundaries

City of Chicago · Community Areas

Polygon shapefile used to spatially join crime incidents to neighborhoods.

How they connect

Listings → **Crimes → spatial join → by area** → **Income → join by area name** → **Final dataset: 6,162 rows, 30 cols**

DATA PREPARATION

4	~7,800	6,162	0
Data Sources	Raw Listings	Final Listings	Nulls in Final Dataset

Focused on short-term rentals only

Removed 1,478 listings with no reviews and a median minimum stay of 32 nights — long-term rentals, not the market we're studying. Also excluded hotel and B&B property types.

Joined crime & income by neighborhood

Each listing matched to one of 77 Chicago community areas. Crime joined via coordinates, income via area name. Zero listings lost in either merge.

Capped extreme prices before modeling

Hotel-style outliers reached $50,000/night. Capped at the 99th percentile ($650/night) so the model reflects the realistic short-term rental market.

Price Distribution

$144	$438	14.2×	$650
Median	Mean	Skewness	Model cap (99th pct)

The $144 median is the right benchmark for a typical host. The mean is pulled up by a small number of luxury outliers.

EDA – Determining variables for model

Criteria

Diversity

There should be enough variance in data points for a variable.

No Multicollinearity

Multicollinearity could confuse the model

Correlation to Price

Variables should be correlated to price

Methods

Value Count

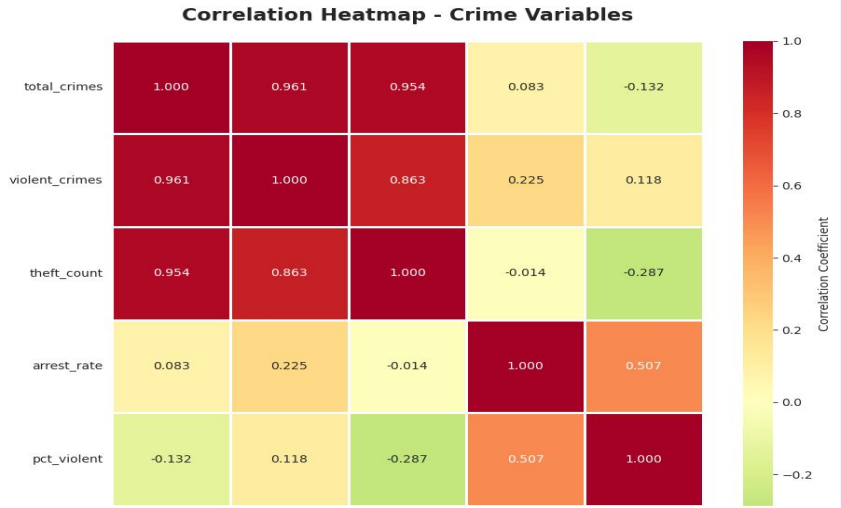
Using the .value_count method to find the count on all unique values

Corr + Heatmap

Finding correlation between each variables (excluding Price)

Corr

Finding correlation between Price and other variables



Variables to Include

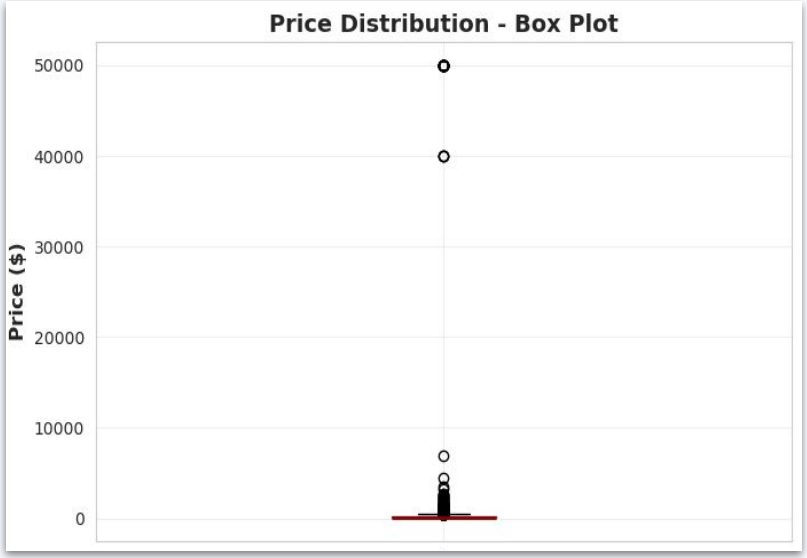
1. **theft count**
   incidents of theft
2. **bathrooms**
   Physical size
3. **is_entire_home**
   Room type flag
4. **accommodates**
   Group capacity
5. **median_income**
   Neighborhood wealth

EDA – Deriving Patterns and Trends on Price

Methods

Groupby + stat measures

Calculating std, mean, median and quartiles on price and variables



Findings

High income + low crime areas underpricing

North Center (median income: $147,701, low crime) charges an average of $182/night, which is around. Areas near downtown charge at a ratio of 3.3-3.4 per $1k of income despite lower income levels.

High Crime Areas Price Higher

Near North Side: 10,394 crimes — highest in Chicago — and avg price $2,558. Urban density beats safety concerns. Central location dominates.

Guests Compare Value Per Person

Price ÷ accommodates was the strongest predictor in both models (coef 88). A $200 listing for 4 people beats a $120 listing for 2 in group travelers' eyes.

MODEL RESULTS

METRIC	V1 — LISTING FEATURES ONLY	V2 — + CRIME & INCOME
Linear R^2	0.4755	**0.5303** ▲
Linear RMSE	$112.02 / night	**$106.01 / night** ▲
Logistic Accuracy	78.95%	**81.57%** ▲
Logistic AUC	0.8764	**0.9004** ▲

TOP LINEAR COEFFICIENTS (V2)

💡 Neighborhood crime & income **improved both models** — but listing characteristics explain most of the variation.

Linear model predicts price within **~$64/night**. Logistic correctly classifies **4 out of 5 listings** as high vs. low price.

KEY FINDINGS & TAKEAWAYS

1 The "Tourist Area" Effect

Higher theft_count areas (Loop, Near North Side) actually command higher prices — tourism density outweighs safety concerns for guests.

2 Capacity is the #1 Driver

Accommodates and bathrooms are the top coefficients in both linear and logistic models — more than location, reviews, or host status.

3 Entire Homes Command Premiums

Entire home listings price meaningfully higher than private or shared rooms — the model assigns a ~$17 premium to entire home listings.

4 Neighborhood Context Adds Value

Adding crime & income improved linear R^2 by 5.5% (0.48 →0.53) and logistic AUC by 0.024 (0.876→0.900) — location matters beyond zip code.

🎯 **Bottom line for hosts & investors:** Prioritize **capacity and property type** over neighborhood prestige. A large entire-home listing in a high-traffic tourist area will consistently outperform a small private room in a safer, quieter neighborhood in nightly revenue.

Limitation: Airbnb listing data is a single snapshot (Sept 22–24, 2025) — seasonal pricing patterns are not captured. ~47% of price variance unexplained; seasonality, listing photos, and description quality are likely missing factors.

LIMITATIONS

Timing Misalignment

Crime: Feb 2025–Feb 2026. Listings: Sept 2025. Income: ACS 2020–2024. The three datasets don't share the same time window — seasonal or annual shifts could affect results.

Moderate Predictive Power

Linear R^2 of 0.53 means ~47% of price variance is unexplained. Key missing signals: listing photos, text quality, dynamic pricing history, and competitor pricing.

Scope of Listings

1,478 long-term rentals and 8 hotel/B&B property types were excluded. The model only applies to active short-term residential listings.

Single City, Single Snapshot

Data reflects September 2025 only — no seasonality captured. Chicago summer vs winter pricing diverges significantly. Results are specific to this market.